

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)		A series of <u>dark lines</u> superimposed on a <u>continuous spectrum</u> (1) [Due to passage of radiation through] <u>atmosphere</u> / <u>photosphere</u> / <u>chromosphere</u> [of star] (1) Accept gases around the star Don't accept stratosphere	2			2		
	(b)		Recall of $I = \frac{L}{4\pi d^2}$ can be implied (1) Correct pair of values from graph (1) Substitution and re-arrange to find L (or P) = 4×10^{26} W or 4×10^{23} kW unit mark (1)	1	1 1		3	2	
	(c)		T calculated from $P = \sigma AT^4$ Substitution and re-arrange ecf on P (1) $T = 5808$ [K] (1) Peak λ identified (490-510 nm) and used in Wien's equation (1) $T = 5800$ [K] (5686-5918 K) – consistent (1) Alternative: Determine T from spectrum graph as above [2 marks] Substitute T into $P = \sigma AT^4$ and re-arrange (1) To confirm either P , A or σ (1)			4	4	4	
			Question 8 total	3	2	4	9	6	0